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Predicting seagrass ecosystem resilience to marine heatwave events of variable duration, frequency and re-occurrence patterns with gaps

Paula Sobenko Hatum [1] | Kathryn McMahon [2] | Kerrie Mengersen [1] | Kieryn Kilminster [3] | Paul Pao-Yen Wu [1]

[1]School of Mathematical Sciences, Science and Engineering Faculty, Queensland University of Technology, Brisbane, Queensland, Australia

[2]School of Science and Centre for Marine Ecosystems Research, Edith Cowan University, Joondalup, Western Australia, Australia

[3]Department of Water and Environmental Regulation, Government of Western Australia, Joondalup, Western Australia, Australia

Correspondence
Paula Sobenko Hatum, School of Mathematical Sciences, Science and Engineering Faculty, Queensland University of Technology, Brisbane, Queensland, Australia.
Email: paula.sobenkohatum@hdr.qut.edu.au

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Abstract

Background: Seagrass, a vital primary producer habitat, is crucial for maintaining high biodiversity and offers numerous ecosystem services globally. The increasing severity and frequency of marine heatwaves, exacerbated by climate change, pose significant risks to seagrass meadows.

Aims: This study acknowledges the uncertainty and variability of marine heatwave scenarios and aims to aid managers and policymakers in understanding simulated responses of seagrass to different durations, frequencies and recurrence gaps of marine heatwaves.

Materials and Methods: Using expert knowledge and observed data, we refined a global Dynamic Bayesian Network (DBN) model for a specific case study on Halophila ovalis in Leschenault Estuary, Australia. The model evaluates the potential impact of marine heatwaves on seagrass resilience, examining stress resistance, recovery and extinction risk.

Results: Simulations of different marine heatwave scenarios reveal significant impacts on seagrass ecosystems. Scenarios ranged from 30- to 90-day heatwaves, with longer durations causing more significant biomass decline, reduced resistance, higher extinction risk and prolonged recovery. For instance, recovery time may increase from 18 to 26 months with four 60-day and from 24 to 47 months with four 90-day marine heatwave events. Increasing the frequency of marine heatwaves from one to four annual events, with no gaps between occurrences, could raise extinction risk from 11% to 55% for 60-day events and from 17% to 83% for 90-day events. However, introducing gaps between heatwaves enhanced resilience, with spaced events showing lower extinction risks and quicker recovery than consecutive yearly events.

Discussion: The study demonstrates the DBN model's utility in simulating the impact of marine heatwaves on seagrass, providing tools for risk-informed assessment of management and restoration efforts. While these simulations align with existing research on temperature impacts on seagrass, they are not empirical.

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Conclusion: Further research is necessary to expand our understanding of climate change effects on seagrass ecosystems, guide policy and develop strategies to strengthen marine ecosystem resilience.

KEYWORDS

climate change, dynamic model, ecological forecasting, extreme climate events, seagrass, thermal stress

1 | INTRODUCTION

Marine heatwaves, defined as areas of unusually high sea surface temperature persisting for days to months, are a significant consequence of anthropogenic climate change (Hobday et al., 2016). These events, driven by atmospheric and oceanographic processes (Schlegel et al., 2017), vary in magnitude, duration and spatial extent (Holbrook et al., 2020). As ocean temperatures rise, marine heatwaves are expected to become more frequent and prolonged (Oliver, Lago, et al., 2018), already causing widespread ecological disruptions, including mortality of marine species (Garrabou et al., 2009; Piatt et al., 2020), shifts in species distribution (Cavole et al., 2016; Hyndes et al., 2017; Wade et al., 2019) and impacts on ecosystem services like fisheries (Mills et al., 2013; Smale et al., 2019).

Seagrass ecosystems, vital to coastal marine biodiversity, are under increasing pressure from climate change disturbances associated with extreme climate events (ECEs) (Smale et al., 2019; Strydom et al., 2020; Webster et al., 2021). In addition to global climate change, their shallow estuarine and coastal habitats make them particularly susceptible to anthropogenic stressors, including coastal development and water quality degradation (Crain et al., 2009; Duarte, 2002; Halpern et al., 2007; Orth et al., 2006; Waycott et al., 2009). The combined effects of these stressors, often interacting, are critical in shaping the current and future states of seagrass ecosystems (Arias-Ortiz et al., 2018; Brown et al., 2014; Darling & Côté, 2008; Grech et al., 2011; Kendrick et al., 2019; Ontoria et al., 2019).

Recent studies attribute the decline in seagrass meadows, in terms of cover and biomass, to extreme heating events (Arias-Ortiz et al., 2018; Rasheed & Unsworth, 2011; Strydom et al., 2020; Waycott et al., 2009). Temperature anomalies as low as 3°C above the typical seasonal averages have led to significant seagrass mortality in temperate and subtropical regions (Marbà & Duarte, 2010; Thomson et al., 2015). The 2003 Mediterranean marine heatwave, characterized by high atmospheric pressure and elevated air temperatures, resulted in seawater temperatures 1–3°C above normal, causing extensive ecological damage, including the loss of seagrass meadows (Garrabou et al., 2009; Marbà & Duarte, 2010). A similar temperature anomaly in 2006 raised concerns about its potential impact on seagrass, though specific consequences are yet to be fully understood (Kersting et al., 2013).

Seagrass species, with their broad temperature tolerance range (0–45°C), have evolved various anatomical and physiological adaptations to environmental fluctuations (Abal et al., 1994; Dawson & Dennison, 1996; Lee et al., 2007). However, these adaptations are insufficient under conditions outside their tolerance limits, such as from heat stress (Bita & Gerats, 2013; Reusch et al., 2008; Yamori et al., 2014). The resilience of seagrass to marine heatwaves, including their ability to resist, recovery following disturbances and capacity for adaptation Wu et al. (2017), Hodgson et al. (2016), Ingrisch & Bahn (2018), is influenced by the frequency, duration and gaps between marine heatwaves (Aoki et al., 2021; Guerrero-Meseguer et al., 2017; Marbà & Duarte, 2010; Olsen et al., 2012; Perkins et al., 2012).

There is a notable gap in understanding how different global warming scenarios will affect marine organisms such as seagrass during marine heatwaves (Frölicher & Laufkötter, 2018; Kendrick et al., 2019; Smith et al., 2022; Unsworth et al., 2014). It is particularly challenging to predict how marine heatwave features—such as duration, frequency and magnitude—will impact survival and population dynamicsnd anticipate potential synergistic effects on seagrass populations. This uncertainty poses challenges in coastal management and necessitates new tools for predicting and mitigating the impacts of marine heatwaves. However, the resilience of seagrass is attributable to the complex interplay of systems within marine ecosystems (Wu et al., 2017). A promising approach is to develop a model to incorporate the multiple associations between species and their environment and combine empirical data with existing knowledge to build scenarios that describe possible alternative futures.

In this study, we propose a Dynamic Bayesian Network (DBN) model to assess the cumulative impact of various stressors on seagrass resilience, focusing on the effects of marine heatwaves. While Bayesian methods are widely used in modelling complex ecological systems and for environmental management (Barton et al., 2012; Borsuk et al., 2004; Pollino et al., 2007), as far as the authors are aware, no work has used a DBN model in predicting seagrass resilience to the effects of marine heatwaves. The overarching goal of this paper is to understand how heat stress arising from marine heatwaves can impact the resilience of seagrass meadows using a dynamic Bayesian modelling approach. We addressed this in three stages: (1) modify and validate an existing DBN for seagrass meadows (Wu et al., 2017) using the guidelines provided by Hatum et al. (2022) to incorporate heat stress nodes using expert elicitation; (2) test the developed methods on a case study; and (3) run scenarios to assess the effect of increasing the

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duration, frequency and return gaps of marine heatwave on the resilience of seagrass defined as resistance, recovery time and risk of extinction (Wu et al., 2017).

2 | CASE STUDY APPLICATION

The Leschenault Estuary in southwestern Australia, spanning 13.5 km in length and 2.5 km in width, is characterized by a shallow platform with varying sediment types. It is connected to the ocean via an artificial channel and is bordered by the Leschenault Peninsula (McComb et al., 2001; Semeniuk et al., 2000). Water temperatures range from 8°C in winter to 28°C in summer (de Lestang et al., 2003). This estuary was chosen as the focus of our case study due to its comprehensive archival data and the existence of an ongoing seagrass monitoring programme. This initiative was revitalized following the significant depletion of seagrass observed during the summer of 2013–2014, leading to the instigation of regular, estuary-wide surveys starting from the summer of 2015. The estuary is home to three seagrass species, with *Halophila ovalis* being the most prevalent. Surveys conducted by the Department of Water and Environmental Regulation have shown that, since 2017 (29.7%), there has been a notable recovery, with seagrass coverage reaching 54.7% of the estuary by 2020 (Bennett et al., 2020). Seagrass loss can be attributed to human-induced stressors, such as dredging or eutrophication, but is also influenced by climate change factors, including altered rainfall patterns, sea level rise and marine heatwaves (Bennett et al., 2020). However, it is crucial to acknowledge that no direct correlation was established between this loss of seagrass and marine heatwave events during the specified timeframe.

In this case study, we focused on fast-growing seagrass species *H. ovalis*, as it was the dominant species at the study sites. Data used in the modelling were provided by the Department of Water and Environmental Regulation, collected as part of their seagrass and water quality programme (Hugues-dit Ciles et al., 2012, Bennett et al., 2020) (for details, see Table S2). Two sites (Sites 1 and 2) from three monitoring sites were chosen because they best represent the distribution range of *H. ovalis* within the estuary, as well as the physical conditions of the estuary (Figure 1) (Thomson & Pages, 2022). The abundance of seagrass, including cover, biomass, light and water temperature, was recorded between 2014 and 2020, with seagrass sampling focused on summer but year-round water quality sampling from 2014 to 2020.

For each site, biomass and seagrass cover measurements were collected once a month during the summer (December–March). Over this time, loggers captured temperature (°C) with a high frequency (20 min sampling rate) and light (photosynthetically active radiation, PAR) at 10-min intervals and integrated into a daily light measurement (moles m^{-2} day^{-1}). To supplement light and temperature for the remainder of the year, data collected fortnightly to monthly in the water quality monitoring programme were used. Temperature was collected with a Hydrolab MiniSonde at each time point at 0.5 m intervals from the surface to the bottom of the water column. In addition, Secchi depth (Z_{SD}) was recorded and was used to estimate light availability for seagrass.

3 | MATERIALS AND METHODS

3.1 | DBN model description

A Bayesian network (BN) is a class of graphical models that illustrates probabilistic relationships among variables using a directed acyclic graph (DAG). Each node in the DAG represents a variable $X_1,X_2,...,X_n \epsilon X$, and directed arcs A indicate dependencies between these variables (Jensen & Nielsen, 2007). Nodes can take various states, such as discrete labels, numerical values or Boolean forms (Bromley et al., 2005). The relationship between a node and its parents is quantified using a Conditional Probability Table (CPT), expressed as $PX = x \mid Par(X)$, where $Par(X)$ is the parent nodes of X and $X = x$ denotes the state of X. The DAG structure implies that each node is conditionally independent of nodes that are not connected. The joint probability distribution of all variables is given by

$$P(X_1,X_2,...,X_n) = \prod_{i=1}^{n} P\{X_i|Par(X_i)\}. \tag{1}$$

Posterior marginal probabilities, $P(X|E)$, are calculated based on observed evidence E (Pearl, 1988). A DBN extends BNs to model dynamic systems or time series data, where a template BN is replicated at discrete points in time, typically with arcs connecting successive time slices (Murphy & Mian, 1999). DBNs are particularly useful for modelling complex systems with cumulative effects and feedback processes. The DBN approach can thus facilitate the prediction of the resilience of a system given the temporal dynamics of the components of the ecosystem and their interactions with natural and anthropogenic stressors (Wu et al., 2017; Wu et al., 2018).

In quantifying resilience, we focus on three key ecological aspects extracted from the literature (Wu et al., 2017). First, resistance (Levin & Lubchenco, 2008; Walker et al., 2004) denotes the ability of an ecosystem to persist even under increasingly frequent and severe pulse disturbances. A system with higher resistance has a greater ability to maintain its state and function under baseline conditions, even after experiencing a disturbance, such as heatwaves. Second, recovery (Halpern et al., 2007) refers to the expected time it takes for an ecosystem to recover following a disturbance. Finally, the risk of local extinction (Holling, 1973) assesses the probability of a system being completely lost over time relative to its baseline conditions. A higher value indicates a greater risk of a complete loss.

The proposed DBN is based on the one developed by Wu et al. (2018), which has 34 nodes, as outlined in Table S1, capturing resilience and ecosystem processes at the scale of a seagrass meadow in monthly time steps. These processes encompass resistance (e.g., physiological measures such as carbohydrate stores that facilitate resistance to pressures up to a point when the carbohydrate

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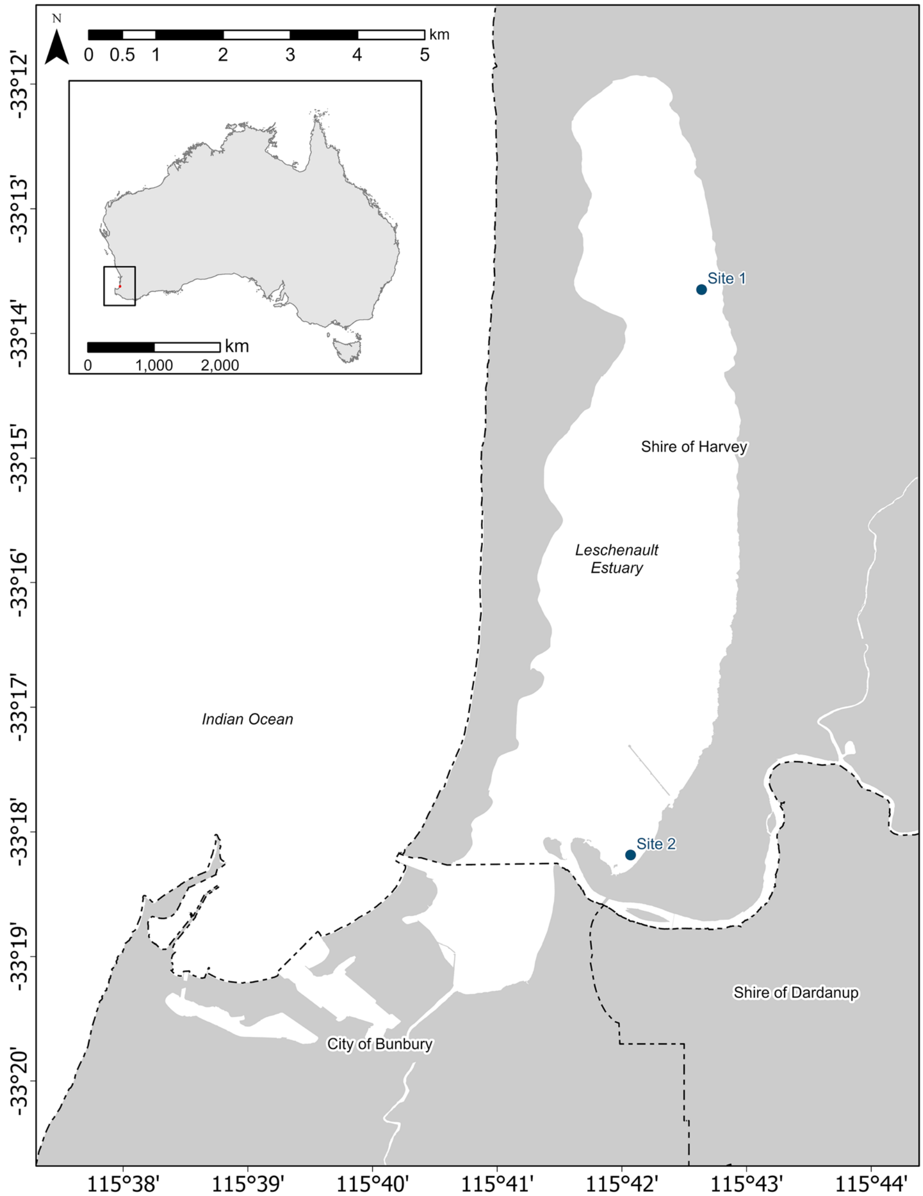

FIGURE 1 General location of Leschenault Estuary in southwestern Australia and the two sampling sites.

stores are depleted), recovery (e.g., reproduction through seeds and growth as production of new leaves, shoots and lateral expansion enables recovery of a meadow), site conditions (e.g., genera present as different genera have different relative carbohydrate stores and growth rates) and environmental factors (e.g., light) (Figure 2, Figure S1a,b,c). However, the model of Wu et al. (2018) does not account for the effects of temperature and heat stress. Therefore, there is a necessity to expand upon it, as discussed in Section 3.2.

3.2 | Adapting the DBN Model for marine heatwave impact in Leschenault Estuary

The structure of the existing model had originally been built and validated by a group of experts (tab. S5 in Wu et al., 2017). Expert elicitation was conducted to identify components of the existing DBN model that needed to be altered to capture the local growth dynamics and seasonal variation of *H. ovalis* meadows located in Leschenault Estuary, including the integration of heat stress as a hazard factor. Consistent with best practices in model adaptation, we focused on modifying the CPTs, redefining nodes and altering node states rather than redesigning the model structure (Grzegorczyk & Husmeier, 2009).

We performed elicitation predominantly with a seagrass biologist (Kathryn McMahon, Edith Cowan University, Australia) and an Estuarine scientist (Kieryn Kilminister, Department of Water and Environmental Regulation). The expert elicitation focused on three main aspects: incorporating temperature and heat stress into the model, discretizing temperature and heat stress nodes and updating the existing DBN model to our case study using guidelines provided by Hatum et al. (2022). This included refining DBN nodes, links and states, drawing on observational data, literature and expert insights for necessary updates to the model's conditional probability tables.

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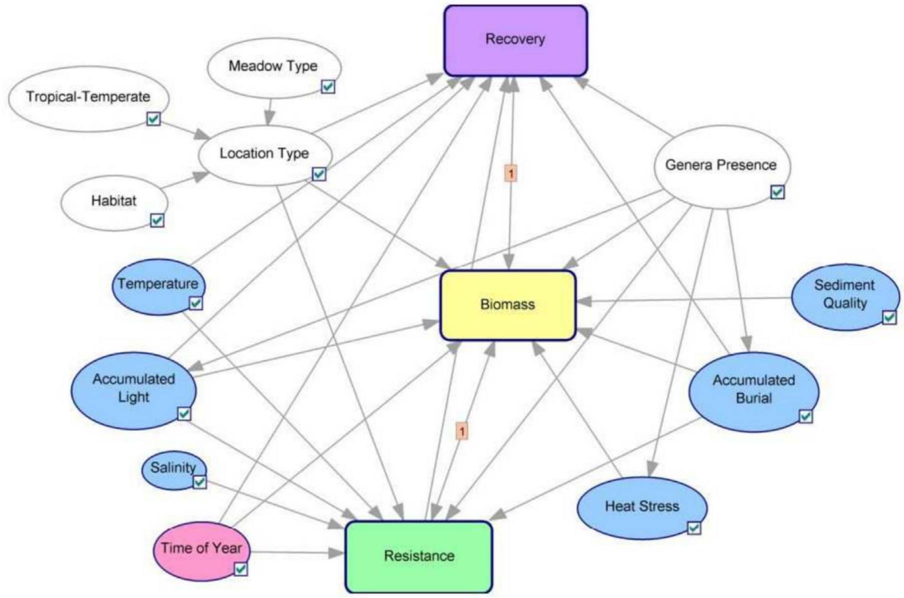

FIGURE 2 The overall Dynamic Bayesian Network (DBN) network structure. Nodes are ovals, and arrows denote causal parent–child relationships in the direction of the arc where a parent node (e.g., meadow type) influences a child node (e.g., location type); conversely, an absence of a link implies conditional independence. Rounded rectangles denote subnetworks, that is, networks that describe a component of the system expressed as a single node in the larger BN (object-oriented approach) (Johnson & Mengersen, 2012). Nodes are coloured as follows: white for site condition nodes, purple for recovery nodes, green for resistance nodes, blue for environmental nodes and yellow for population (biomass) nodes. Time of year (pink) represents the months of the year from January through December. Figure adapted from Wu et al. (2017), Nature Communications 8:1263, supporting material, fig. 7. The figure was developed using GeNIe, a development platform for creating graphical decision-theoretic models. The following symbology is used in this figure: a node with a check mark: a node is ticked when inference has been successfully executed. A double-headed arrow indicates that there are arcs heading both ways between two subnetworks. For example, a double-headed arrow between a node N and a subnetwork S means that there is at least one node in S that depends on N and that there is at least one node in S that influences N. An arrow labelled with a [1] indicates a connection to a subsequent time slice $(t+1)$.

3.2.1 | Temperature and heat stress nodes

Seagrasses under moderate heat stress experience carbon imbalance due to a proportionally higher increase in respiration than photosynthesis and undergo irreversible damage to their photosynthetic apparatus when stress exceeds specific thresholds (Lee et al., 2007; Marsh et al., 1986; Staehr & Borum, 2011). Two different mechanisms by which temperature affects seagrass were captured in the model: optimal temperature and heat stress. The former relates to the optimal temperature range for plant growth and potential adverse effects on the physiological status of plants that impede their ability to function normally when the temperature is outside that range. Additionally, elevated temperatures can disrupt the recruitment process from seeds since earlier stages of the life cycle of plants are more vulnerable to warming compared to mature adults (Duarte et al., 2018). In contrast, heat stress captures mortality or loss in shoot density as a result of elevated temperatures. Consequently, a temperature node was linked to physiological status and seed recruitment rate, and a heat stress node was linked to a loss in biomass to incorporate seagrass mortality.

As our formulation is for discrete DBN, each of the nodes in the DBN is discretized into states, and by doing that, we can capture decision thresholds, such as biological thresholds relating to plant function or decision thresholds relating to management (Wu et al., 2017). Seagrass growth dynamics are influenced by seasonal patterns and species-specific temperature ranges (Campbell et al., 2006; Lee et al., 2007; Marbà & Duarte, 2010). Photosynthesis in aquatic plants, being temperature dependent, increases with temperature up to an optimum point, after which it declines at higher temperatures (Campbell et al., 2006; Marsh et al., 1986; Staehr & Borum, 2011). Temperature tolerance varies among seagrass species: tropical species can withstand higher temperatures (Adams et al., 2017; Collier et al., 2017) compared to subtropical and temperate species (Marsh et al., 1986; Masini et al., 1995; Pedersen et al., 2016).

Temperature states were defined as a temperature range based on the published thermal optima of the particular genera/species. For *H. ovalis* in the Leschenault Estuary, the optimal temperature range was set at 17°C to 30°C, based on two studies. The upper limit was informed by Ontoria et al. (2020), who found maximum gross photosynthesis for *H. ovalis* at 31°C in southwestern Australia. The lower limit came from Said et al. (2021), who observed high photosynthetic rates at 17°C in a study conducted 100 km from the Leschenault Estuary. Therefore, the sub-optimal temperature state was outside of this range, both above and below.

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TABLE 1 The description of the two nodes added to the DBN to account for temperature and the heat stress induced by marine heatwave.

Node	State	Description
Temperature	Optimal	The temperature is within the optimal temperature range (17–30°C).
	Sub-optimal	Temperature is above/below the optimal temperature range, affecting the physiological status of plants and recruitment.
Heat stress	Effect	There is an effect of a marine heatwave event. The thermal optima is exceeded, leading to plant mortality.
	No effect	There is no effect of a marine heatwave event.

Heat stress in a given month was classified as 'effect' (100% heat stress) or 'no effect' (0% heat stress), based on whether or not a heat stress event had occurred in that month. Temperature states for each month were defined as optimal and suboptimal (Table 1). The effect of heat stress is used to capture the risk of seagrass mortality during marine heatwaves, which is accounted for by assuming the temperature thresholds for seagrass mortality have been exceeded for a given month(s). The discretization of the heat stress node into 'effect' and 'no effect' states is conceptually grounded in the thermal limit (T_{lim}), as outlined in Marbà et al. (2022), which signifies the temperature threshold triggering seagrass die-off during heat waves.

Baseline environmental conditions, defined by the probability of above saturation light and the optimal temperature for each month, serve as a reference for normal fluctuations and a metric for identifying the onset and impact of marine heatwaves. When a marine heatwave occurs, the heat stress node is adjusted to 'effect' to indicate heat stress onset. This adjustment reflects the increase in the likelihood of biomass loss and a shift in temperature categorization to suboptimal, indicating an environment that could potentially harm plant functionality and the recruitment of younger stages. By considering heat stress, we increase the likelihood of biomass loss.

3.2.2 | Updating CPTs for site-specific application

Having added a node for temperature and one for heat stress, the CPTs of these nodes need to be populated along with updates to the CPTs of all their child nodes and other relevant nodes (Hatum et al., 2022). In general, the CPTs must be adapted to reflect the growth dynamics of the *H. ovalis* population in Leschenault Estuary, as well as the seasonal variation in their reproduction and heat stress. The CPTs of the following nodes were updated from expert elicitation: physiological status of plants, nodes used to capture seagrass growth dynamics (baseline shoot density and loss in shoot density) and seasonal variation in seagrass reproduction (seed density and recruitment rate from seeds) (Table S4).

A comprehensive framework for elicitation, based on the guidelines of Hatum et al. (2022) and Wu et al. (2017), was employed to update the CPTs. This framework used linguistic labels and scenario-based methods to elicit the conditional probabilities to quantify the DBN. The elicitation process took the form of scenarios (or stories), an intuitive way for experts to make sense of the evidence (Pennington & Hastie, 1993; Uusitalo, 2007) and linguistic labels of certainty (extremely likely, very likely, likely, 50/50, unlikely, very unlikely, extremely unlikely and impossible). To elicit the expert's conditional probabilities for each node of interest, questions were phrased as follows: 'If seagrasses were under good light conditions but exhibited a weak ability to resist a hazard, what is the probability of the plants being impacted by the effect of heat stress?'. Using CPTs, the model captures both the uncertainty of outcomes (due to the unpredictable nature of ecological interactions and environmental changes) and the variation in the associations between species and their environment.

3.2.3 | Model validation

The model underwent partial validation by comparing its predictions with out-of-sample data on biomass, light and temperature from the Leschenault Estuary. Biomass data collected in the Leschenault Estuary were used to validate the prediction of the model. We focused on total biomass, encompassing both above and below-ground biomass, measured in cores. This measurement indicates that it is specifically assessed in areas where seagrass is present. Core sampling in the Leschenault Estuary involved collecting cylindrical samples to assess seagrass density and health, measuring various plant components and converting these measurements to a per-metre squared basis.

However, considering that the seagrass meadows were not continuous, and we intended to account for the entire meadow area, we also incorporated seagrass cover as an additional metric. This approach was adopted because biomass measurements based on cores were stratified to seagrass patches, whereas the cover estimates were derived from randomly placed quadrats at each site. Coverage assessment was based on the percentage of area occupied by seagrass in these samples, providing insights into the spatial distribution and abundance of seagrass in the estuary. Thus, the monitoring data were adjusted for a more reliable estimate of seagrass biomass at the meadow scale. The biomass measured from each core (g DW m^2) at each site and time was multiplied by the average cover estimates from each site at the respective time, enhancing the accuracy of the biomass estimates.

The model predicts biomass levels categorized as high, moderate, low and zero biomass. These categories can be directly aligned with management thresholds, such as those outlined in the water quality framework for a moderate level of protection (Anzecc, 2000). However, this alignment necessitates a transformation from

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continuous biomass measurements. To accomplish this, we adopt the approach described by (Wu et al., 2018), utilizing a hierarchical ordinal regression model:

$$g(y_{i,t}) = \beta_{0,i} + \beta_{1,i}\sin\left(\frac{t}{6\pi}\right) + \beta_{2.i,Site}. \tag{2}$$

The coefficients of the model are in Table S5. Under this formulation, $g^{-1}(y_{i,t})$ represents the probability of state i (high, moderate, low and zero) of biomass at time t (month of the year). The regression has coefficients β_0 and β_1, which are the global intercept and slope for the seasonal effect of months t, respectively, and coefficient β_2, which is the random effect used to capture the differences between sites. Model (1) was fitted using Hamiltonian Monte Carlo (HMC) with the R package brms (Bürkner, 2018), employing default flat priors (Supporting Figures S4 and S5 and Tables S6 and S7). This transformation facilitated the comparison of the biomass state probabilities obtained from the model with the observed data collected in the study area.

As the data were not used in training the model, we employed mean-squared error (MSE) as a metric to validate the model, comparing the predicted posterior marginal distribution with the observed biomass distributions (Wu et al., 2018). Additionally, an independent expert from the Department of Water and Environmental Regulation responsible for managing the Leschenault Estuary was consulted to appraise the BN itself, the model outputs for no marine heatwave and different configurations of marine heatwave scenarios. The DBN model's predictions of seagrass biomass were based on inputs such as genera, climate (tropical or temperate), depth, tidal exposure (subtidal or intertidal) and meadow type (transitory or enduring). Additionally, the model considered the probability of light exceeding saturation levels, optimal temperature and effect of heat stress.

3.3 | Marine heatwave scenarios

Once the temperature and heat stress nodes were incorporated into the DBN model and the validation was satisfactory, the resilience of seagrass in the case-study model was assessed against a range of marine heatwave scenarios. The model operates on a monthly timescale, simulating the seagrass ecosystem's response over a 10-year period, starting from established baseline conditions. These baselines, including the probability of above-saturation light and optimal temperature for each month, were derived from historical data (as detailed in Table S3). The probability of above saturation light per month, $\delta\left(x^{light}_{\text{above sat}}, t\right)$, $t = \{Jan, Feb, \ldots, Dec\}$ was estimated from the proportion of days with sufficient hours of above saturation light. The probability of optimal temperature was computed similarly, $\delta\left(x^{temperature}_{optimal}, t\right)$, $t = \{Jan, Feb, \ldots, Dec\}$ (Table S3).

In our study, we utilized existing literature to justify the selection of marine heatwave scenarios for our DBN model. Different combinations of duration, frequency and re-occurrence with gaps of heatwaves were tested, as both duration and frequency are predicted to increase under climate change (Darmaraki et al., 2019; Frölicher et al., 2018; Hobday et al., 2016). For instance, studies indicate marine heatwaves occur one to three times annually, with durations ranging from 40 to over 250 days in certain regions. Predictions suggest more frequent, extensive and prolonged marine heatwaves with climate change (Oliver, Donat, et al., 2018; Sen Gupta et al., 2020).

Our scenarios, reflecting these trends, included various combinations of heatwave duration (30, 60 and 90 days) and frequency (one to four events in a decade), with and without gaps between events in a 10-year period. The scenarios started in January, aligning with the Southern Hemisphere summer when marine heatwaves, are most intense (Sen Gupta et al., 2020). This approach allowed us to simulate the most severe marine heatwave impacts on marine ecosystems. As not all studies contain all the aspects included in the formulation of the scenarios (duration, frequency and gap among events), the chosen scenarios were based on multiple global and local studies on marine heatwave events, ensuring they were representative of real-world conditions (Darmaraki et al., 2019; Frölicher et al., 2018; Jacox et al., 2022; Oliver, Donat, et al., 2018; Sen Gupta et al., 2020; Strydom et al., 2020).

A total of 30 scenarios were designed. Broadly, these heat event scenarios can be divided into three groups (Figure 3): Group 1, a single marine heatwave event that increases in duration from 1–3 months; Group 2, which is multiple marine heatwave events of increasing frequency for each duration, for example, 2 years in a row up to 4 years in a row; and Group 3, a combination of two to four marine heatwave events, with 30, 60 and 90 days duration, with gaps of 1 to 2 years between events. Note that the probability of above saturation light, the genera/species and location-specific characteristics such as climate (tropical or temperate), depth and tidal exposure (subtidal or intertidal) and transitory or enduring (permanent) type of meadow were all key inputs for each site and assumed to be equal across all scenarios.

The outputs of the DBN model, which comprise probability distributions for every node over every time point (month), were assessed over three different time periods as per Wu et al. (2018): (1) the 12-month baseline time period, T_B, during which no heat events are simulated; (2) the hazard, T_H, the time period from 1 to 10 years in which one or more heatwave is simulated; and (3) the consequence time period T_C, when the marine heatwave is over but seagrass may still be affected. The resilience of seagrass to heat stresses induced by marine heatwave was assessed by three metrics: resistance, risk of extinction and recovery are evaluated using the following Wu et al. (2018).

1. Resistance assesses the condition of the seagrass immediately after the last heat event, before any recovery. This is compared to the seagrass state in the same month of the baseline period to evaluate the immediate impact of the heatwave.

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FIGURE 3 Marine heatwave scenarios. Marine heatwave scenarios are constructed by varying three main factors: duration of a single event, increased duration and frequency of multiple events occurring annually without gaps and the re-occurrence patterns with gaps between marine heatwave events. The yellow colour indicates events having a duration of 30 days, while the orange and red colours denote durations of 60 and 90 days, respectively.

2. Risk of extinction is estimated as the accumulated probability of zero population during T_C compared to that during the corresponding baseline time period, that is, $P(X_{biomass}(T_C) = zero\ state|E)$. For example, if $T_C = \{Jan - Year_1, Feb - Year_1, \ldots, Jan - Year_3\}$, then we set $T_B = Jan, Feb, \ldots, Dec, Jan, \ldots, Dec, Jan$.
3. Recovery differs from the risk of extinction in that we are interested in the entire probability distribution over time, not just the zero state. Measures the time taken for the seagrass to return to and consistently maintain levels within 20% of its baseline population during T_C. This metric helps understand the duration and effectiveness of the recovery process post-heatwave.

Simulations were conducted for each scenario, yielding state probability trajectories over time and associated metrics of resilience (resistance, recovery and risk of extinction) (see Figures 4–8).

4 | RESULTS

The model validation results are presented in Section 4.1, employing the methods outlined in Section 3.2. Furthermore, the outcomes of the analysis of marine heatwave scenarios are detailed in Section 4.2, utilizing the methodologies described in Section 3.3.

4.1 | Marine heatwave DBN model validation

In the case study presented, an ordinal regression model (Section 3.2) was employed to transform observed biomass data into state probabilities of biomass for *H. ovalis* at the two chosen sites. The DBN's model predictive performance in terms of biomass was quantified using the MSE metric, which compared the predicted state probabilities to the observed data. The MSE for Sites 1 and 2 was 0.06, suggesting strong predictive validity of the DBN and is similar to the MSE reported in the global study in Wu et al. (2017). The relative size of this MSE, which was found to be close to 1 (0.99), indicates that the predictions from the model align with the natural variability of the data. However, as the data for Site 2 were not as reliable due to the placement of temperature and light loggers far away from the biomass monitoring location, Site 1 was chosen for scenario analysis. A visualization of biomass predictions in terms of probability and a weighted mean is provided in Figure S2a,b.

4.2 | Marine heatwave scenarios

The DBN model estimated the impact of the marine heatwave scenarios (Section 3.3) and its associated stressors on the resilience of seagrass in terms of recovery, resistance (Figure S3a) and risk

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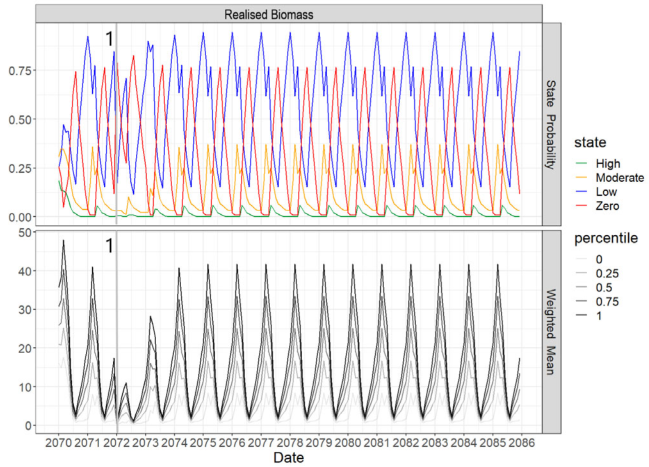
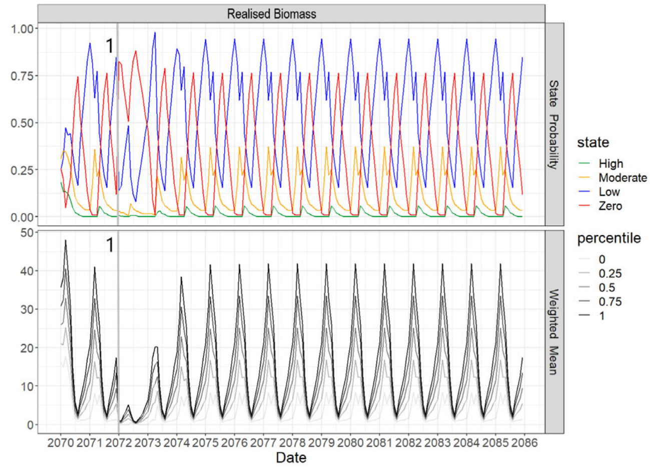
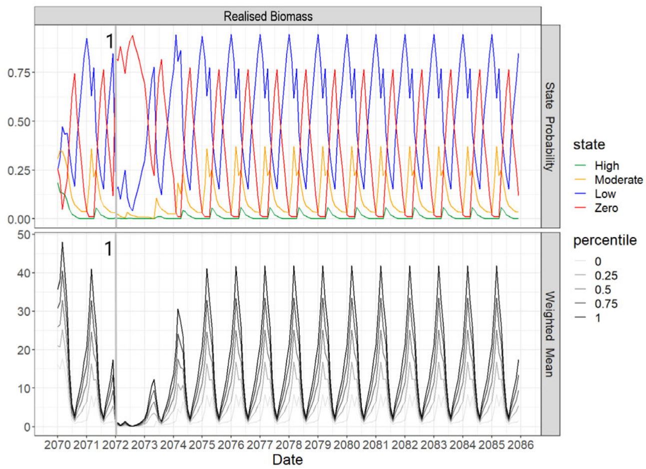


(a) 30 days single event



(b) 60 days single event



(c) 90 days single event

FIGURE 4 The model predicted-state probabilities for biomass for *H. ovalis* located at Site 1. The initial 24 months are used for initialization to allow the system to enter the baseline pattern. The top plots are the probability of each biomass state, and the bottom plots show the weighted mean of the expected value and the interquartile range. Scenarios considering a unique marine heatwave. Each event differs in duration. (a) Scenario 1: 30 days duration, (b) Scenario 5: 60 days duration, and (c) Scenario 9: 90 days duration.

of extinction (Figure S3b) criteria (Table 2). In general, the most significant changes in the resilience of seagrass to various heat stress scenarios were found to be associated with recovery time and risk of extinction. In contrast, resistance was low in all scenarios, with slight variation across the scenarios. Increasing the duration of marine heatwaves from 30–90 days had the greatest impact on seagrass resilience with lower resistance, longer recovery times and a greater risk of extinction the longer the heatwave persisted. Increasing the frequency or number of events had a similar response, especially with a longer duration of marine heatwaves. When there was a gap of 1–2 years between recurring heatwaves, this had a positive effect on resilience, depending on the duration and number of marine heatwaves. The results from the three broad groups of scenarios (duration with a single event, increasing frequency of marine heatwaves and the time between marine heatwaves) are explored further in the following sections.

4.2.1 | Increased marine heatwave duration with a single event

The duration of the heatwave, from 30 to 90 days, had a substantive adverse effect on the abundance and resilience of the seagrass meadow with a single marine heatwave event. As the duration of the marine heatwave increased, the decline in seagrass biomass was greater and remained lower for longer (Figure 4). This was reflected in the resilience metrics, where resistance or the ratio of the population after a heatwave decreases as the duration of the heat events increases. Resistance reduced from 0.055 (5.5%) for 30 days to 0.033 (3.3%) for 60 days and 0.009 (0.9%) for 90 days. Recovery time also increased from 15 months for 30 days, 18 months for 60 days and up to 24 months for 90 days. The cumulative risk of extinction only increased slightly with an increased duration of marine heatwave from 1.07 (0.7%) for a 30-day event to 1.11 (11%) for a 60-day event, up to 1.17 (17%) for a 90-day event.

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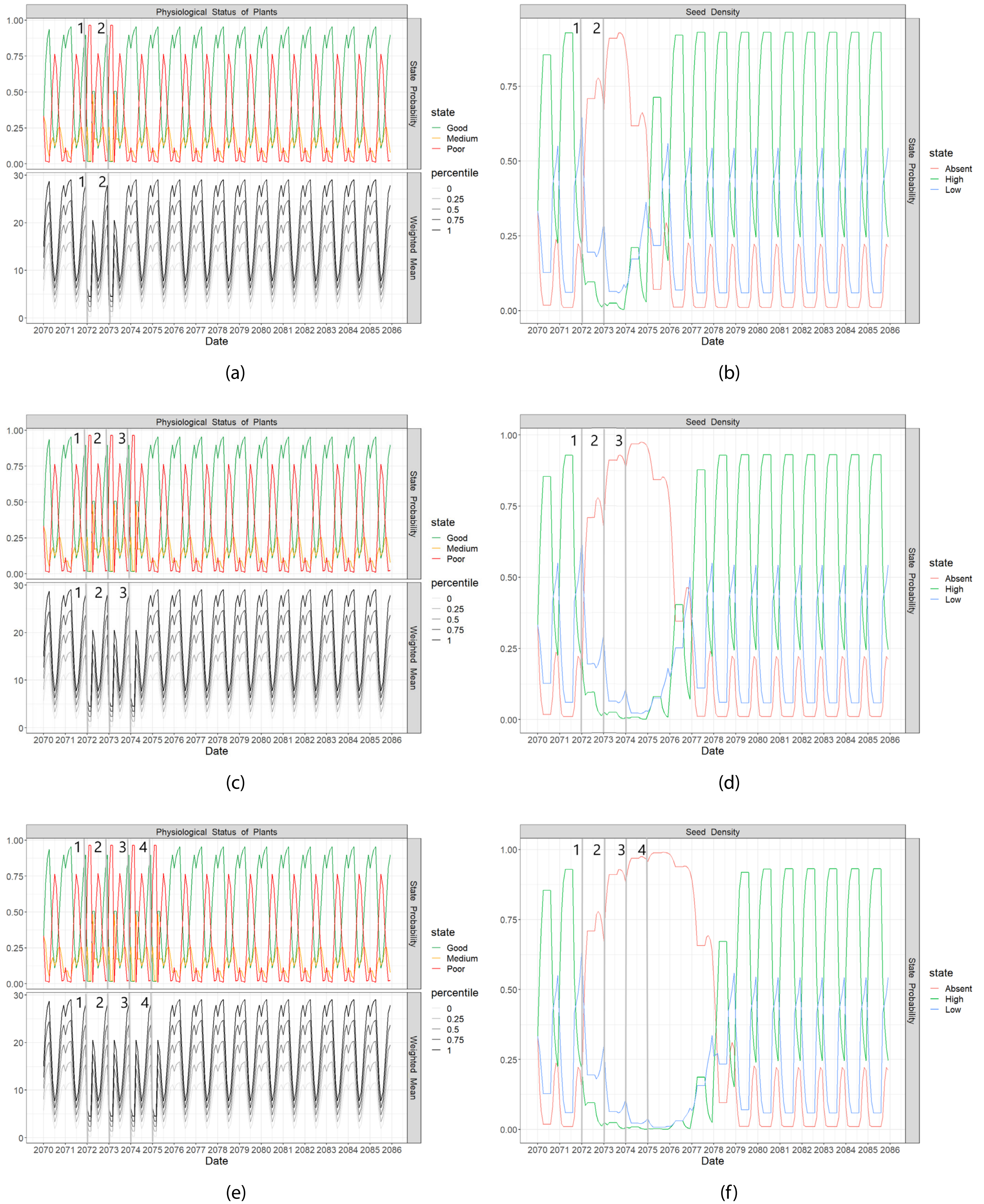

FIGURE 5 The model predicted-state probabilities for the physiological status of plants and seed density for *H. ovalis* located at Site 1. The initial 24 months are used for initialisation to allow the system to enter the baseline pattern. The top plots are the probability of each biomass state, and the bottom plots show the weighted mean of the expected value and the interquartile range. Scenarios of marine heatwaves that occur annually and in consecutive years. All scenarios have 90 days duration and frequency of occurrences varying from 1 to 4 events within a decade. (a) Physiological status of plants and (b) seed density for Scenario 10, two consecutive events; (c) physiological status of plants and (d) seed density for Scenario 11, three consecutive events; (e) physiological status of plants and (f) seed density for Scenario 12, four consecutive events.

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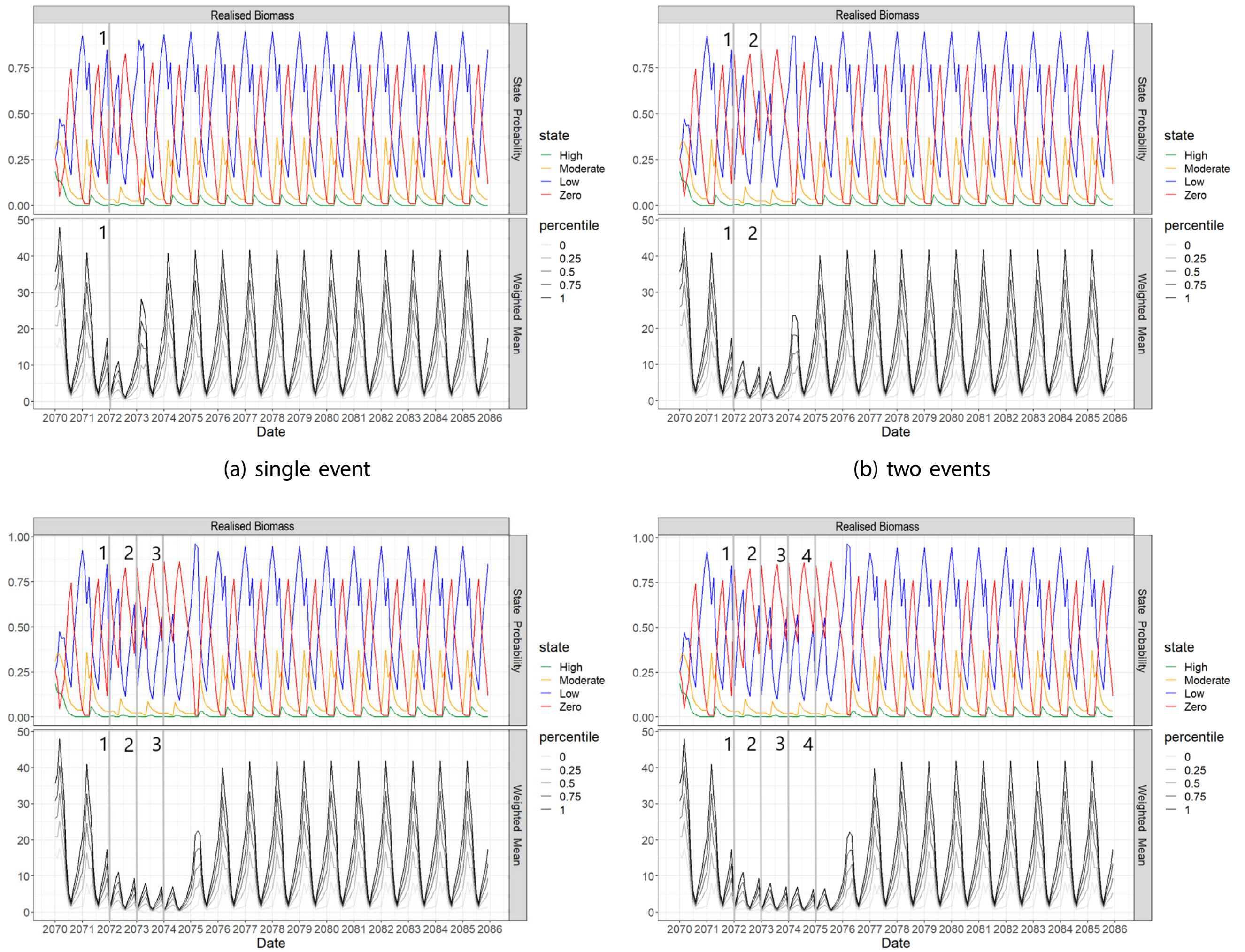

(a) single event (b) two events

(c) three events (d) four events

FIGURE 6 The model predicted-state probabilities for biomass for *H. ovalis* located at Site 1. The initial 24 months are used for initialization to allow the system to enter the baseline pattern. The top plots are the probability of each biomass state, and the bottom plots show the weighted mean of the expected value and the interquartile range. Scenarios of marine heatwaves occur annually and in consecutive years. All scenarios have a duration of 30 days, and the frequency of occurrences varies from one to four events within a decade. (a) Scenario 1: single event, (b) Scenario 2: events with recurrence of 2 years, (c) Scenario 3: events with recurrence of 3 years, and (d) Scenario 4: events with recurrence of 4 years.

4.2.2 | Increased marine heatwave duration and frequency for multiple events

Recurrent heatwaves had a negative impact on the physiological health of plants, with an increased probability of poor physiological status and a reduced probability of high physiological status. There was also a greater probability of no seed bank, and this probability increased with the number of events (Figure 5). For a 30-day heatwave, the resilience metrics that were most affected by increasing the frequency of consecutive events were resistance, with a steady decline from 5.5% of the baseline after one event down to 2.95% of the baseline after four events (Figure 6), and risk of extinction, with an increase of 23.9% (from 1.066 after one event to 1.305 after four events). The recovery time was very similar, 15–16 months, regardless of whether it was a single event or four recurrent events.

Increasing the duration of successive heatwave occurrences from 30 to 60 days had a substantial impact on resistance and recovery. As the number of consecutive 60-day heatwaves increased, the resistance decreased from 0.033 (3.3%) after one event to zero after four events. Recovery also rose from 18 months for one event to 26 months for four events. The risk of extinction increased by 43.8% (from 1.111 after one event to 1.549 after four events). For consecutive events of 90 days, resistance to the heatwave was reduced with increasing frequency, reaching zero already after two recurrent occurrences. Recovery time was longer, and this increased with the number of heat events, for example, 24 months with a single occurrence up to 47 months with four recurrent events. By

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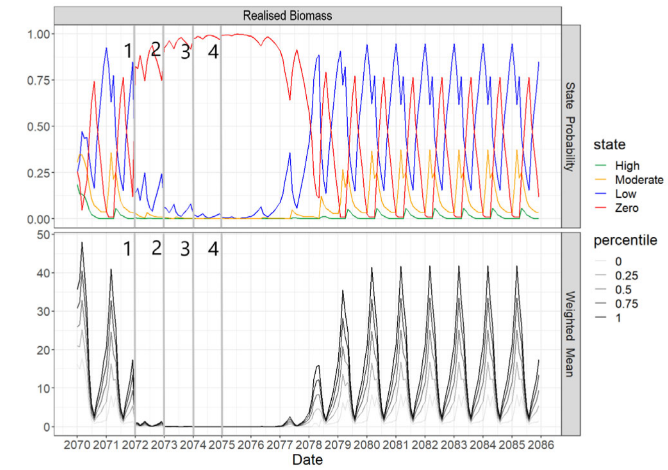
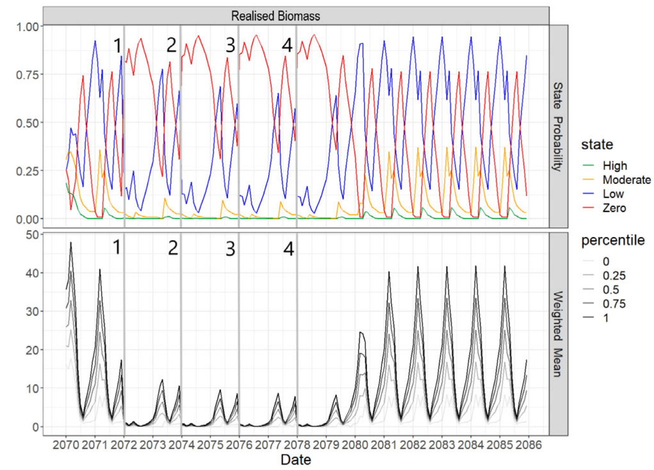
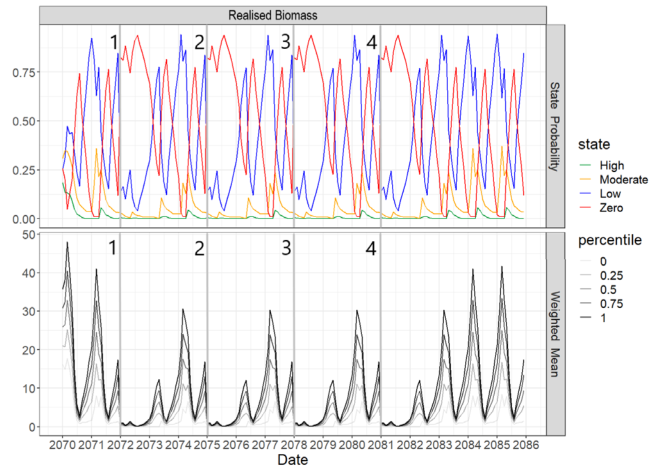


(a) one event per year



(b) one event every two years



(c) one event every three years

FIGURE 7 The model predicted-state probabilities for biomass for *H. ovalis* located at Site 1. The initial 24 months are used for initialisation to allow the system to enter the baseline pattern. The top plots are the probability of each biomass state, and the bottom plots show the weighted mean of the expected value and the interquartile range. Scenarios consider four events of marine heatwaves, with a duration of 90 days and different gaps of occurrence among the marine heatwave events. (a) Scenario 12: no gap, (b) Scenario 21: gap of 1 year, and (c) Scenario 30: gap of 2 years.

increasing the number of 90-day events fourfold, the risk of extinction increased by 66% (from 1.170 after one event to 1.833 after four events).

4.2.3 | Gaps between marine heatwave events

In terms of recovery, the decline in biomass across multiple heat stress events was less severe when there were gaps between recurrent heatwaves (Figure 7). When there was only a one-year gap between successive heatwaves, the recovery time was reduced in all scenarios. Except for three to four marine heatwaves of 90-day duration, the recovery returned to what was observed in a single event (Figure 8). For example, adding 2 years between events reduced recovery from 26 to 18 months for four recurrent events of 60 days, and adding 2 years between events reduced recovery from 47 to 24 months for four recurrent events of 90 days. For recovery for four recurrent events, with 30, 60 and 90 days duration, when a 2-year gap was added between events, the recovery time was the same as a single event with the same duration.

Regarding resistance, when a 2-year gap was added between events, the resistance was comparable to a single event. Regardless of the number of occurrences (2–4), there was no discernible difference in resistance. Heat stress had less effect on resistance when the four events were spaced further apart. In Scenarios 21 and 30, for example, the gap between events was 1 and 2 years, and the resistance of the seagrass was 0.41% and 0.86% of the baseline, respectively.

Adding gaps of one to 2 years in relation to the frequency of events resulted in a decrease in the risk of extinction; however, as the

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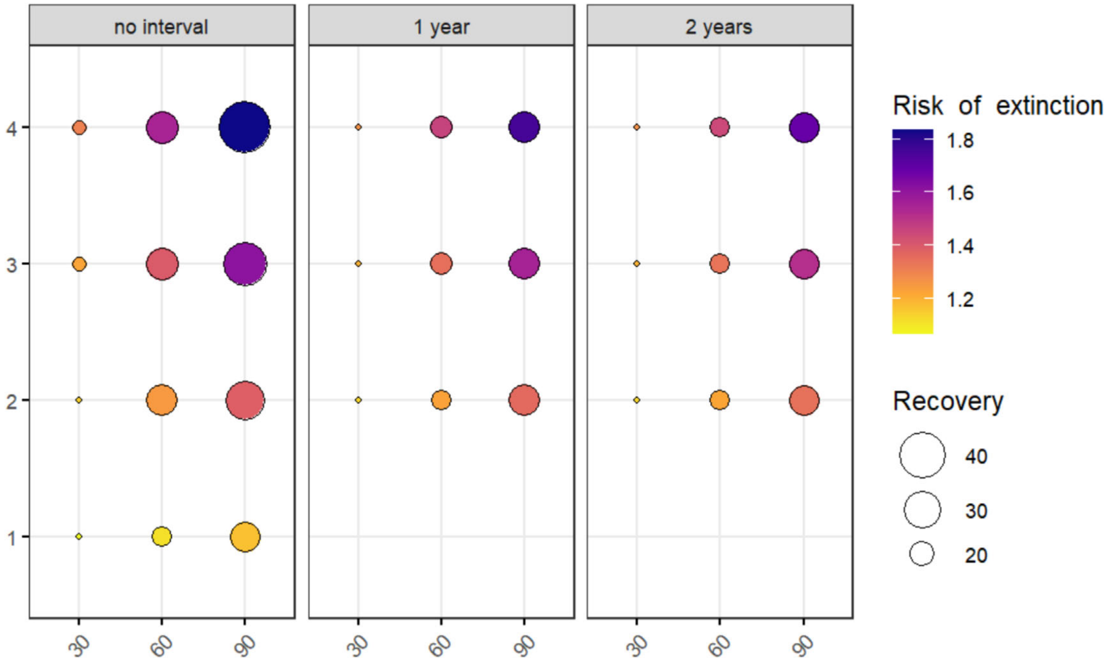

FIGURE 8 Recovery time (measured in months) and risk of extinction are assessed across various marine heatwave scenarios. Recovery refers to the time taken to return within 20% of baseline condition after a marine heatwave, while the risk of extinction represents the likelihood of seagrass meadow reaching zero compared to baseline, considering various heatwave scenarios. These scenarios differ in duration (measured in days), frequency (ranging from one to four annual events) and occurrence interval (1 year, 2 years and no gap) over a decade. Colours range from yellow to purple, with yellow indicating a lower risk of extinction time and purple representing a higher risk of extinction. Bubble size indicates the recovery time, with the smallest bubbles representing a low recovery time and the largest bubbles indicating a high recovery time.

range of occurrence increased, the effect of heat stress on seagrass risk of extinction reduced. For example, for events of the same duration and frequency, such as Scenarios 12 (four recurrent events with no gaps between occurrences), 21 and 30 (four events with gaps of recurrence of 1 and 2 years, respectively), the estimated risk of extinction was reduced by 8.15% and 14.72% when comparing Scenario 12 (1.8332) with Scenarios 21 (1.7517) and 30 (1.6860), respectively.

5 | DISCUSSION

ECEs can result in significant ecological change (Babcock et al., 2019). As ECEs are predicted to become more frequent and intense, understanding the direction and magnitude of ecological responses to these events is necessary for predicting their effects on an ecosystem. Since seagrass ecosystems are intrinsically dynamic, responding to environmental change across various scales and predicting species' response to ecosystem changes is challenging (Kendrick et al., 2019; O'Brien et al., 2018). Although mathematical models to assess seagrass resilience to climate change have been developed, these models tend to account for only a small number of processes and often fail to incorporate an understanding of how multiple stressors affect resilience non-linearly and synergistically (Aoki et al., 2021; Carr et al., 2012; Fourqurean & Rutten, 2004). As a result, there is a gap in bringing together the whole-of-systems for the seagrass ecosystem response to climate change to support management, especially with respect to MHW.

This study introduces a powerful tool designed to assess ecological responses in dynamic seagrass ecosystems, particularly to marine heatwaves, which are anticipated to become more frequent and severe as a result of climate change. Our focus was on assessing seagrass resilience to marine heatwave scenarios by extending a validated DBN model (Wu et al., 2017), addressing a gap in comprehensive models that consider multiple environmental stressors and their synergistic effects on seagrass dynamics. The model, with a predictive MSE of 0.06, encompasses various processes, including both growth and physiological responses to environmental factors, as well as their cumulative interactions over time, which lead to emergent behaviours in complex systems (Holling, 2001). Its design incorporates a probabilistic method that adeptly captures, predicts and analyses these complex interactions. Although this study utilized specific thresholds based on literature for marine heatwaves, the framework can be easily adapted to simulate other environmental thresholds, as suggested by Hatum et al. (2022). Using laboratory or field experiments, seagrass responses to heatwaves of varying frequency and duration, for example, may be assessed (Guerrero-Meseguer et al., 2017; Marín-Guirao et al., 2016; Olsen et al., 2012). However, because of the time and effort involved, it is not feasible to experimentally reproduce different combinations of heat events with gaps between them; instead, models must be used.

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TABLE 2 Heat stress results for scenarios in (Figure 3) varying by duration (days), frequency and interval of occurrence within a decade.

Scenarios	Duration	01	02	03	04	05	06	07	08	09	10	Recovery	Resistance	risk of extinction
Scenario 1	30											15	0.0550	1.0661
Scenario 2	30											15	0.0386	1.1431
Scenario 3	30											16	0.0321	1.2231
Scenario 4	30											16	0.0295	1.3046
Scenario 5	60											18	0.0330	1.1113
Scenario 6	60											25	0.0099	1.2490
Scenario 7	60											26	0.0043	1.3987
Scenario 8	60											26	0	1.5495
Scenario 9	90											24	0.0089	1.1705
Scenario 10	90											32	0	1.3838
Scenario 11	90											37	0	1.6117
Scenario 12	90											47	0	1.8332
Scenario 13	30											15	0.0534	1.1331
Scenario 14	30											15	0.0534	1.2002
Scenario 15	30											15	0.0534	1.2672
Scenario 16	60											18	0.0285	1.2279
Scenario 17	60											19	0.0277	1.3454
Scenario 18	60											19	0.0276	1.4631
Scenario 19	90											25	0.0054	1.3602
Scenario 20	90											25	0.0044	1.5550
Scenario 21	90											25	0.0041	1.7517
Scenario 22	30											15	0.0548	1.1322
Scenario 23	30											15	0.0548	1.1983
Scenario 24	30											15	0.0548	1.2644
Scenario 25	60											18	0.0328	1.2229
Scenario 26	60											18	0.0328	1.3345
Scenario 27	60											18	0.0328	1.4460
Scenario 28	90											24	0.0086	1.3423
Scenario 29	90											24	0.0086	1.5141
Scenario 30	90											24	0.0086	1.6860

Note: Recovery (months), resistance and risk of extinction estimation, considering different marine heatwave scenarios. Over a decade, scenarios have varied in terms of duration (days), frequency and interval of occurrence.

Our case study found that *H. ovalis* meadows exhibit varying resilience to increased frequency and duration of marine heatwave, with significant declines in resilience for heat stress events exceeding 30 days. These findings reflect those observed in real-world heatwaves lasting 5 days, 7–21 days, 28 days, and more than a month (Buñuel et al., 2021; Díaz-Almela et al., 2009; Guerrero-Meseguer et al., 2017; Hendriks et al., 2017; Jordà et al., 2012; Marbà & Duarte, 2010; Marín-Guirao et al., 2016; Olsen et al., 2012; Traboni et al., 2018). Aligning with previous research (Aoki et al., 2021; Perkins et al., 2012), our findings suggest that the frequency of marine heatwave events is a critical factor in determining seagrass recovery capacity, with each successive event escalating the risk of loss. However, our study advances this understanding by quantifying the impacts in terms of recovery, resistance and risk of extinction, thereby providing a more comprehensive perspective on the resilience of seagrass meadows under various heat stress conditions.

Our study highlights that the duration of heat stress significantly influences recovery times in marine environments, more than the frequency of heatwave events. Recovery periods ranged from 1.5 years following a single 60-day event to nearly 4 years for four consecutive 90-day events within a decade. This suggests that prolonged exposure to high temperatures plays a critical role in dictating recovery trajectories, aligning with findings from Nguyen et al. (2020) regarding *Halophila stipulacea* under extended thermal stress. Conversely, shorter heatwave events of 30 days resulted in quicker recoveries, typically under 2 years, indicating a threshold duration beyond which ecological resilience is compromised. Our longest observed recovery period, nearly 4 years, underscores the severe impact of repeated, prolonged stress, aligning with Serrano et al. (2021) on the potential for incomplete ecosystem recovery from continuous thermal stresses.

In Shark Bay, Western Australia, research on *Halodule uninervis* revealed its robust recovery abilities amidst marine heatwaves and reduced light conditions Jung et al. (2023). Subjected to predicted future warming temperatures, the species maintained a stable leaf metabolome throughout the heatwave and recovery phase, highlighting its adaptive mechanisms to withstand high temperatures and dim lighting. Colonizing genera, such as *H. ovalis*, invest

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extensively in seed production and vegetative growth, which contribute to its resilience, but frequent marine heatwave events risk seed bank depletion (Scenarios 10–12) and reduced biomass (Scenarios 1–4), as shown in our Figures 5 and 6. An example of recovery through seed banks is observed in Hervey Bay, Queensland, where floods and turbid conditions led to a significant reduction of deep-water Halophila, approximately by 1,000 km^2. Remarkably, a near-complete recovery was noted within 6–8 years (Kilminster et al., 2015).

The physiological status of the plants, depicted in Figure 5, consistently deteriorates with each successive marine heatwave event. However, it shows a swift recovery when optimal temperature conditions are restored. This pattern of decline and recovery in plant physiology correlates with the changes in biomass, as shown in Figure 6, where a resurgence in physiological status coincides with biomass restoration (Gunderson et al., 2016). Unlike the relatively resilient physiological responses, seed density does not show a similar recovery, instead escalating in its decline as heatwave events become more frequent. This indicates a potential for seed bank depletion over time, which could lead to reduced rates of return to baseline conditions (Gladstone-Gallagher et al., 2024). The trend towards decreasing biomass with increased frequency of marine heatwave events further supports this observation, emphasizing the prolonged impact on the ecosystem's structural integrity.

While *Halophila* meadows survived heat stress, they exhibited low resistance across all scenarios (York et al., 2015). Longer and more frequent marine heatwave events led to greater biomass loss, a trend also noted by Duarte et al. (2018). Repeated 60- to 90-day marine heatwave events significantly increased the extinction risk for *Halophila* meadows, particularly when occurring three to four times per decade. The risk of extinction rose sharply, from 11% after one such event to 55% after four, underscoring their heightened vulnerability. Remarkably, this elevated risk persisted even with a 2-year interval between events. These observations underscore the urgent necessity for strategies that bolster the resilience of ecosystems facing frequent and prolonged stressors (Gunderson, 2000).

The recovery time of seagrass meadows from disturbances, including marine heatwave events, is influenced by various factors beyond those considered in this study, such as diminished light, pollution and sedimentation. These additional stressors, along with high temperatures, can exacerbate declines and impede recovery. For instance, the interaction of eutrophication with seawater warming could significantly affect the abundance and distribution of *Posidonia oceanica* under different climate change scenarios (Johnson & Mengersen, 2012). Our current understanding of these combined impacts is limited, necessitating further research to explore these synergies, particularly in climate-sensitive seagrass species. Recovery times can be prolonged, especially when meadows face multiple stressors like dredging, sedimentation, eutrophication and habitat fragmentation. Consequently, our estimates of recovery times post-heat stress might be underestimated. Future studies should delve into the mechanisms of these synergistic effects, which are likely to intensify with rising global temperatures.

5.1 | Implications for Conservation

The DBN model serves as a valuable tool for managers in two key areas: risk assessment and mitigation planning, and policy development and advocacy. First, the model enables the evaluation of potential risks to seagrass ecosystems by simulating a broad spectrum of marine heatwave scenarios, extending beyond the specific cases examined within this research. By simulating these scenarios, managers can identify high-risk areas and allocate resources more effectively. This aspect of the model is crucial for developing strategies to mitigate risks, such as enhancing the resilience of seagrass meadows or reducing additional stressors like water pollution. Furthermore, the architecture of the model permits its extension to assess the impacts of a diverse range of environmental stressors, thereby underscoring its utility and adaptability in comprehensive ecosystem management strategies. Second, the model plays a significant role in policy development and advocacy. It provides empirical evidence that can support policy decisions, advocating for stronger seagrass protections. As new data on seagrass responses to climate change become available, the model can be updated to refine predictions, ensuring that management and policy decisions are based on the most current and accurate information. This adaptability makes the DBN model an indispensable tool for informed decision-making in seagrass ecosystem management.

The DBN model developed by (Wu et al., 2018) initially aimed to predict the resilience of seagrass meadows to dredging disturbances, focusing specifically on three distinct genera and locations in Australia: *Amphibolis* in Jurien Bay, *Halophila* in Hay Point and *Zostera* in Pelican Banks, Gladstone. Demonstrating its broad applicability and scalability, this general model was subsequently employed to assess dredging impacts on seagrass resilience across 28 global locations (Wu et al., 2017). This extension from local to global scales underscores the model's capability to generalize across different seagrass genera and ecological sites. Furthermore, guidelines proposed by (Hatum et al., 2022) for adapting the DBN model—encompassing phases of knowledge acquisition, revision and design, and site application—have facilitated the model's adaptation for diverse ecological domains. The modifications implemented in this paper demonstrate the applicability of the model to assess the resilience of seagrass ecosystems to different heatwave scenarios. The versatility of the model will enable it to be applied in other environmental settings. As an example, the model could be used to predict seagrass response to marine heatwaves for species with slower recovery rates, such as *Amphibolis antarctica* and *Posidonia australis* Strydom et al. (2020), Kendrick et al. (2019).

Our study highlights the adaptability and relevance of the DBN model approach in comprehending and managing seagrass ecosystems amidst climate change pressures. Nonetheless, the precise measurement and interpretation of resilience, particularly in scenarios of diverse loss and recovery trajectories, pose ongoing complexities. Enhancements to the model, including integration with weather, hydrodynamic and socio-ecological systems, are imperative for a more holistic approach to ecosystem management and climate

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scenario planning. Additionally, the periodic climatic phenomena of La Niña and El Niño play a crucial role, altering the frequency and intensity of ECEs (Glantz, 2001). These cycles can lead to periods of high heat stress or recovery gaps, potentially introducing further stressors like flooding, which impacts light availability. Recognizing the timing and impact of these cycles is vital for ecosystem managers and policymakers, facilitating informed decisions that reflect the current resistance capacity and anticipated stressor events. This understanding is instrumental in crafting adaptive management strategies that account for the broad array of environmental variables affecting seagrass meadows. Moreover, the DBN accounts for and can be extended to study other combinations of stressors, such as the impact of heat and light arising from flooding turbidity and more complex scenarios, thereby providing a flexible framework for simulating and predicting ecosystem responses under various environmental conditions.

In general, this study reaffirms the adverse effects of increasing global temperatures and deteriorating water quality on seagrass meadows, underscoring the critical need for adaptive management strategies to strengthen the resilience of these essential ecosystems.

6 | CONCLUSION

Simulations of marine heatwave scenarios showed that longer and more frequent heatwaves severely impact seagrass ecosystems, causing greater biomass loss, reduced resistance and increased extinction risk, with longer recovery times. Recovery could lengthen from 18 to 26 months for four 60-day marine heatwaves and from 24 to 47 months for four 90-day marine heatwaves. Extinction risk may rise from 11% to 55% for 60-day events and from 17% to 83% for 90-day events as marine heatwave frequency increases. Introducing 1- or 2-year gaps between events, however, leads to improved resilience, lower extinction risks and quicker recovery compared to back-to-back yearly events.

Our findings contribute significantly to the global effort to conserve, monitor, manage and restore seagrass ecosystems, providing a foundation for informed policy discussions and the development of effective climate change responses. However, since we focused on the heat stress caused by marine heatwaves, we assumed that all other conditions were optimal while developing our scenarios. Therefore, it is essential to recognize that recovery may vary when meadows are subjected to the impact of other multiple stressors.

AUTHOR CONTRIBUTIONS

Paula Sobenko Hatum led the writing of the manuscript. Paula Sobenko Hatum and Paul Pao-Yen Wu designed the model and the computational framework and analysed the data. Paul Pao-Yen Wu, Kerrie Mengersen and Kathryn McMahon were involved in planning and supervising the work. Kathryn McMahon provided expert knowledge and ecological analysis. Kieryn Kilminster collected the data. All authors provided critical feedback and helped shape the research, analysis and manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no competing interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in Tables S3 and S5 (Dryad Digital Repository: https://doi.org/10.5061/dryad.rr4xgxdf3).

STATEMENT ON INCLUSION

In our study, we were privileged to collaborate with authors from various countries and regions within Australia, including scientists from the area under investigation. These contributors brought invaluable local insights. By involving these authors from the beginning of our research, we ensured that the diverse perspectives they offered played a crucial role in shaping the study's design. Whenever relevant, literature published by scientists from the region was also cited; efforts were made to consider relevant work published in the local language.

ORCID

Paula Sobenko Hatum https://orcid.org/0000-0003-3640-2522

Paul Pao-Yen Wu https://orcid.org/0000-0001-5960-8203

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